

ESCUELA POLITÉCNICA SUPERIOR

FLUID MECHANICS I PROBLEM 2 (10 POINTS).

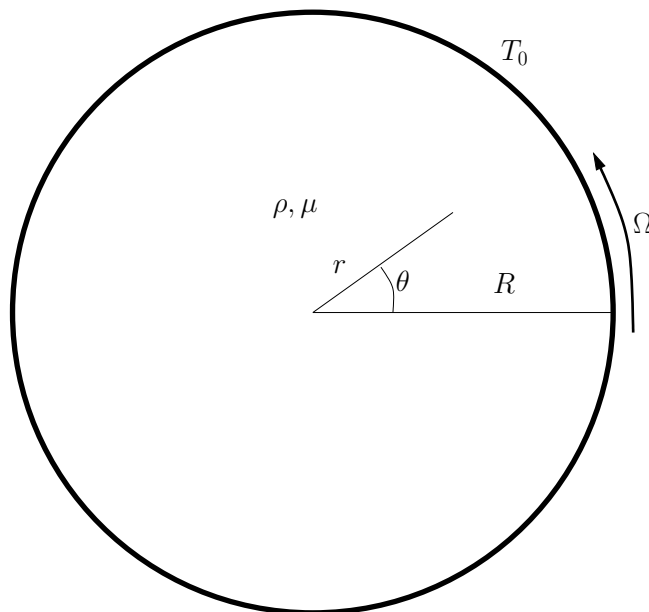
21-06-2022
Time: 60 minutes

NAME:.....

GRADE:.....

The sketch shows a circular cavity full of liquid of density ρ and viscosity μ surrounded by an outer case at temperature T_0 . The fluid is initially at rest, temperature T_0 and pressure p_0 . At $t = 0^+$, the outer case is impulsively set to rotate with a constant angular velocity Ω .

1. Use the Π theorem to reduce the parametric dependence of the torque M (per unit length) necessary to sustain the constant angular velocity of the outer case from $t = 0$ to $t \rightarrow \infty$.
2. Formulate the **full** mathematical problem that determines the evolution of the fluid field (velocity, pressure and temperature) in the cavity $0 \leq r \leq R$ for $0 < t < \infty$, including boundary and initial conditions. Consider the pressure in the fluid adjacent to the wall is known to be p_1 . Simplify the problem as much as possible to obtain the differential equations, with initial and boundary conditions, that determine the azimuthal velocity v_θ , T and p . Viscous dissipation might not be negligible.
3. Find v_θ , T and p in the case $t \rightarrow \infty$.
4. From the last result, obtain the adequate scales v_{θ_c} , r_c , respectively for v_θ , r . Find the adequate time scale t_c using order-of-magnitude analysis in the time-dependent differential equation for v_θ . Make the problem non-dimensional introducing the scales in the dimensional problem.
5. Define the expression that needs to be evaluated to find the torque M at any instant of time t , using the detailed solution of the non-dimensional problem, scaling M adequately.
6. Compare the last result with that of part #1 and comment the result



Note: It is strongly recommended to use cylindrical coordinates

① $M = M(t, s, \mu, R, R)$

	M	L	T
M	1	1	-2
t	0	0	1
s	1	-3	0
μ	1	-1	-1
R	0	0	-1
R	0	1	0

TORQUE PER UNIT LENGTH

$\det 1 \neq 0 \Rightarrow \mu, R, R$ ARE D.I

PI THEOREM: $5-3 = 2$ PARAMETERS

$$\pi_0 = \frac{M}{\mu R^2}$$

$$\pi_1 = \Omega t$$

$$\pi_2 = \frac{s R^2}{\mu}$$

$$\rightarrow \pi_0 = f(\pi_1, \pi_2)$$

② CONTINUITY: $\frac{1}{r} \frac{\partial}{\partial r}(r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} v_\theta = 0$ (1) $\rightarrow r v_r = f(t) = 0 \Rightarrow v_r = 0$
 $\downarrow$
 0 SYMMETRY
 $r=R: v_r=0$

MOMENTUM-R: $\frac{\partial}{\partial t} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_r) \right) + \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial \theta}(r v_\theta) \right) = -\frac{\partial p}{\partial r} + \mu \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_r) \right)$ (2)

MOMENTUM- θ : $\frac{\partial}{\partial t} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_\theta) \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_r) \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_\theta) \right)$ (3)

ENERGY: $\frac{\partial}{\partial t} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_r) \right) + \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial \theta}(r v_\theta) \right) = \phi_r + \frac{k}{r} \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_r) \right)$ (4)
 $\phi_r = \mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \right]^2$

SIMPLIFYING:

$$\begin{aligned} v_r &= 0 & (1') \\ \frac{1}{r} \frac{\partial}{\partial r}(r v_\theta) &= \frac{\partial p}{\partial r} & (2') \\ \frac{\partial v_\theta}{\partial t} &= \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_\theta) \right) & (3') \\ \rho c \frac{\partial T}{\partial t} &= \frac{k}{r} \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_r) \right) + \mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \right]^2 & (4') \end{aligned}$$

I.C: $t=0: v_\theta=0; T=T_0$
 B.C: $r=R: v_\theta=\Omega R; T=T_0$
 $r=0: v_\theta=0; \frac{\partial T}{\partial r}=0$

③ $t \rightarrow \infty \Rightarrow \frac{\partial}{\partial t} = 0$: (3') $\Rightarrow \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_\theta) \right) = 0 \rightarrow \frac{1}{r} \frac{\partial}{\partial r}(r v_\theta) = A \rightarrow r v_\theta = \frac{A r^2}{2} + B$
 $v_\theta = \frac{A r}{2} + \frac{B}{r}$
 $\downarrow$
 NO SINGULARITY
 $r=R: v_\theta=\Omega R$
 $\Rightarrow v_\theta = \Omega r$

(2') $\Rightarrow \frac{s \Omega^2 r^2}{r} = \frac{\partial p}{\partial r} \Rightarrow p = s \Omega^2 \frac{r^2}{2} + C \rightarrow P(R)=P_1$
 $P = P_1 + \frac{1}{2} s \Omega^2 R^2 \left[\left(\frac{r}{R} \right)^2 - 1 \right]$

(4') $\Rightarrow \phi_r = 0 \Rightarrow \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r}(r v_r) \right) = 0 \rightarrow r \frac{\partial T}{\partial r} = D \rightarrow T = \frac{D}{r} + E \rightarrow T = T_0$
 $\downarrow$
 NO SINGULARITY

$$\textcircled{4} \left. \begin{aligned} V_0 &= \Omega R \\ \tau_c &= R \end{aligned} \right\}$$

$$\frac{\partial V_0}{\partial t} = \nu \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r V_0) \right)$$

$$\frac{V_0}{t_c} \sim \nu \frac{V_0}{R^2} \rightarrow t_c \sim \frac{R^2}{\nu}$$

USING THESE SCALES:

$$V = \frac{V_0}{\Omega R}$$

$$z = \frac{t}{R^2/\nu}$$

$$\eta = \frac{r}{R}$$

$$\left. \begin{aligned} V &= \frac{V_0}{\Omega R} \\ z &= \frac{t}{R^2/\nu} \\ \eta &= \frac{r}{R} \end{aligned} \right\} \rightarrow \frac{\partial V}{\partial z} = \frac{\partial}{\partial \eta} \left[\frac{1}{\eta} \frac{\partial}{\partial \eta} (\eta V) \right]$$

$$z=0: V=0$$

$$z>0: \eta=1: V=1$$

$$\eta=0: V=0$$

PROBLEM WITHOUT PARAMETERS

THE SAME CAN BE DONE WITH T:

$$\frac{T-T_0}{T_0} = \theta \rightarrow \frac{\nu c}{R^2} T_0 \frac{\partial \theta}{\partial z} = \frac{k T_0}{R^2} \frac{1}{\eta} \frac{\partial}{\partial \eta} \left[\eta \frac{\partial \theta}{\partial \eta} \right] + \frac{\mu \Omega^2 R^2}{R^2} \left[\eta \frac{\partial}{\partial \eta} \left(\frac{V}{\eta} \right) \right]^2$$

$$\frac{\partial \theta}{\partial z} = \frac{Pr}{\eta} \frac{\partial}{\partial \eta} \left[\eta \frac{\partial \theta}{\partial \eta} \right] + \frac{\Omega^2 R^2}{c T_0} \left[\eta \frac{\partial}{\partial \eta} \left(\frac{V}{\eta} \right) \right]^2$$

$$z=0: \theta=0$$

$$z>0: \eta=1: \theta=0$$

$$\eta=0: \frac{\partial \theta}{\partial \eta}=0$$

$$Pr = \frac{\mu c}{k}$$

$$\frac{\Omega^2 R^2}{c T_0}: \text{NON-DIMENSIONAL PARAMETER}$$

$$\textcircled{5} M = \tau_{\theta} \Big|_{r=R} \cdot R \cdot 2\pi R \quad ; \quad \tau_{\theta} = \mu \left[r \frac{\partial}{\partial r} \left(\frac{V_0}{r} \right) + \nu \frac{\partial \theta}{\partial z} \right] = \frac{\mu \Omega R}{R} \eta \frac{\partial}{\partial \eta} \left(\frac{V}{\eta} \right)$$

$$M = 2\mu \Omega R^2 \pi \underbrace{\left[\eta \frac{\partial}{\partial \eta} \left(\frac{V}{\eta} \right) \right]_{\eta=1}}_{f(z)} \Rightarrow \frac{M}{2\pi \mu \Omega R^2} = f(z)$$

$$\text{NOTICE THAT } z \rightarrow \infty \Rightarrow V \rightarrow 0 \Rightarrow f(z \rightarrow \infty) \rightarrow 0 \Rightarrow \frac{M}{2\pi \mu \Omega R^2} \rightarrow 0$$

$$\textcircled{6} \text{ COMPARING WITH \#1: } \pi_0 = f\left(\frac{t\nu}{R^2}\right) \quad \pi_1' = \frac{\pi_1}{\pi_2} \quad \text{CONSISTENT.}$$